
Rethinking Contrastive Loss in CLIP Post-training: A Complementary Framework with Frozen Text Encoder

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Abstract

CLIP serves as a foundational vision-language model and the de facto vision encoder for downstream VLMs such as LLaVA. Post-training offers a lightweight route to refine CLIP, but recent work argues that the standard contrastive loss is unsuitable for post-training due to catastrophic forgetting under small batches, motivating designs that abandon the contrastive objective in favor of distillation. We revisit this premise and find that, for the InfoNCE objective, the reported forgetting is driven primarily not by insufficient negatives but by an inappropriate magnitude of the contrastive temperature τ : with τ set sufficiently small, contrastive post-training improves rather than degrades the pretrained CLIP, which we explain through the temperature dependence of the InfoNCE gradient. Building on this finding, we propose **ComCLIP**, a lightweight single-epoch post-training recipe that freezes CLIP’s text encoder—so the refined vision encoder is a drop-in replacement with unchanged architecture and inference cost—and trains the vision encoder with a properly-tempered contrastive loss, an MSE anchoring loss against the original CLIP, and a relational distillation loss from DINOv2. Over multiple seeds, ComCLIP matches the self-distillation baseline CLIP-Refine on zero-shot classification while significantly improving the transferability of visual features, measured by linear probing (48.99 vs. 42.28 on ViT-B/16), and on ViT-L/14 it also improves MMVP over CLIP-Refine (24.20 vs. 19.01); CLIP-Refine remains stronger on image-text retrieval. Used as a drop-in vision encoder for LLaVA-1.5-7B without realigning the projector or LLM, ComCLIP yields no net change across 8 VLM benchmarks, i.e., the refinement does not break downstream compatibility. Code and models are available at <https://github.com/showstarpro/ComCLIP.git>.

1 Introduction

Vision-language foundation models, particularly CLIP (Radford et al., 2021), serve as the de facto vision encoder for modern vision-language models (VLMs) such as LLaVA (Liu et al., 2024a). Despite its success, CLIP exhibits residual issues including the modality gap (Liang et al., 2022) and systematic visual blindness on fine-grained pairs (Tong et al., 2024), which propagate into downstream VLMs. Re-training CLIP from scratch (Zhai et al., 2023; Sun et al., 2023) is prohibitively expensive and breaks compatibility with the existing CLIP-based ecosystem. *Post-training*—refining a pretrained CLIP on a small dataset with limited compute—offers a pragmatic alternative. We target a developer who upgrades the vision encoder of a CLIP-based VLM without retraining its projector or LLM: the refined encoder must stay in the original CLIP image-text space, and the visual features the VLM consumes should become more transferable.

Existing CLIP post-training methods fall into two categories: self-distillation methods such as CLIP-Refine (Yamaguchi et al., 2025) that use image-text soft labels, and external-teacher distillation

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methods such as KUEA (Gong et al., 2025) that align the geometric structure of CLIP’s visual representations with DINOv2’s (Oquab et al., 2024) via kernel-based methods. Notably, both lines deliberately *avoid* the standard image-text contrastive loss—the very objective that produced CLIP. CLIP-Refine, for instance, argues that contrastive post-training is fundamentally limited by the small batch sizes affordable in this regime, which provide insufficient negative samples and lead to catastrophic forgetting; they report that even scaling the batch size up to 2,048—near the practical ceiling for post-training—still fails to recover the zero-shot performance of the original CLIP. This pessimistic conclusion about contrastive loss has shaped subsequent designs that rely on distillation rather than direct contrastive learning.

Finding 1: Contrastive temperature is a major underappreciated factor in CLIP post-training.

We revisit this premise and find that, for the InfoNCE objective, the reported forgetting is driven primarily not by insufficient negatives in small batches but by an inappropriate magnitude of the contrastive temperature τ . Directly inheriting the pretraining initialization $\tau \approx 0.07$ (i.e., $1/\tau \approx 14$) leads to performance degradation, whereas a substantially smaller $\tau = 0.01$ (i.e., $1/\tau = 100$) does not. The mechanism is visible in the InfoNCE gradient: the gradient on a negative pair decays as $\exp(-(s_{ii} - s_{ij})/\tau)$ in its margin to the positive. Because a pretrained CLIP already separates matched from mismatched pairs, a small τ makes the gradient on easy pairs vanish, so the pretrained geometry is left largely intact, and concentrates the update on the few hard negatives, which refines it (Sec. 3.3). On ViT-B/16 (CC3M, 1 epoch, batch size 1,024), naive contrastive post-training with $\tau = 0.07$ drops 12-dataset zero-shot accuracy from 61.82 to 57.90, reproducing prior reports; with $\tau = 0.01$, the same objective improves zero-shot accuracy to 62.90 under the same compute budget.

Finding 2: Different post-training objectives act on separable metric families. While properly-tempered contrastive loss alone is beneficial, it leaves the transferability of visual features, measured by linear probing, essentially unchanged. Our loss-decoupling ablation shows that the objectives we consider act on largely separable groups of metrics: the contrastive loss drives zero-shot classification and retrieval, anchoring to the pretrained encoder preserves transferable visual features, and relational distillation from DINOv2 adds further linear-probe transferability. We do not claim this is a unique or exhaustive decomposition of post-training failure modes; it is the empirical basis for combining the objectives. Building on these findings, we propose **ComCLIP**, a lightweight single-epoch post-training recipe that freezes the text encoder—so the refined vision encoder is a drop-in replacement with unchanged architecture and inference cost—and trains the vision encoder under three losses that follow one principle, *refine relative structure while anchoring absolute geometry*: a properly-tempered contrastive loss for alignment, an MSE anchoring loss against the original CLIP, and a relational distillation loss from DINOv2.

On ViT-B/16 and ViT-L/14 with one epoch on CC3M or COCO Caption, ComCLIP and the self-distillation baseline CLIP-Refine sit at different points of a trade-off. Over three seeds on ViT-B/16, the two are at parity on 12-dataset zero-shot accuracy (63.00 ± 0.13 vs. 62.87 ± 0.11 , not significant), CLIP-Refine is ahead on image-text retrieval, and ComCLIP is substantially ahead on linear probing (48.99 ± 0.35 vs. 42.28 ± 0.62 , $p < 0.001$), an advantage that also holds when both methods use the same post-training data. On ViT-L/14, ComCLIP additionally improves MMVP over CLIP-Refine trained on the same data (24.20 ± 0.86 vs. 19.01 ± 1.86). Swapped into LLaVA-1.5-7B without any re-alignment, ComCLIP yields no net change across 8 VLM benchmarks: the refinement preserves downstream compatibility rather than improving downstream accuracy.

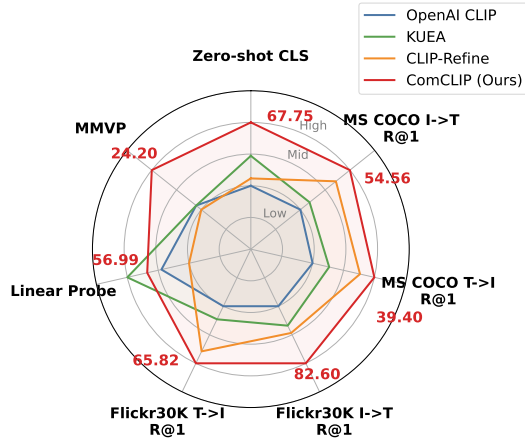


Figure 1: Multi-dimensional comparison on ViT-L/14 (values from Table 1). Relative to the original OpenAI CLIP, ComCLIP improves all four evaluation dimensions (zero-shot classification, retrieval, linear probing, MMVP), whereas prior post-training methods exhibit characteristic trade-offs. Each axis is min-max normalized per metric; ComCLIP’s MMVP is the 3-seed mean (Sec. 4.3).

Contributions. (1) We show that, for InfoNCE, the failure of contrastive post-training reported in prior work is driven primarily by the magnitude of τ rather than by batch size, isolate this effect from batch size and temperature learnability, and explain it through the temperature dependence of the InfoNCE gradient. (2) We propose **ComCLIP**, a lightweight single-epoch post-training recipe with a frozen text encoder that combines a properly-tempered contrastive loss with anchoring and relational distillation. (3) With multi-seed statistics, we show that ComCLIP significantly improves linear-probe transferability over CLIP-Refine at zero-shot parity on ViT-B/16 and ViT-L/14, improves MMVP over CLIP-Refine on ViT-L/14, and serves as a drop-in vision encoder for LLaVA-1.5-7B without degrading downstream performance.

2 Related Work

CLIP and its limitations. CLIP (Radford et al., 2021) pioneered web-scale contrastive image-text pretraining and serves as the de facto vision encoder for modern VLMs (Liu et al., 2024a). While subsequent work improved pretraining via better objectives (SigLIP (Zhai et al., 2023)), rewritten or cleaner data (LaCLIP (Fan et al., 2023)), more efficient training recipes (CLIPA (Li et al., 2023a)), and stronger architectures (EVA-CLIP (Sun et al., 2023)), CLIP still exhibits the modality gap (Liang et al., 2022) and visual blindness on fine-grained pairs (Tong et al., 2024), both of which propagate into downstream VLMs (Li et al., 2024). These pretraining studies also document that the temperature interacts with batch size and training recipe, which provides context for our Finding 1; our setting differs in that the model starts from a converged CLIP and sees only a small corpus for a single epoch.

Freezing one tower. LiT (Zhai et al., 2022) freezes a pretrained *image* encoder and tunes the text encoder to it. ComCLIP is the mirror image: the *text* encoder is frozen and the image encoder is refined, so that the refined encoder remains compatible with every system built on the original CLIP text space (e.g., LLaVA’s projector).

CLIP post-training. Post-training refines a pretrained CLIP with limited compute. **Self-distillation** methods such as CLIP-Refine (Yamaguchi et al., 2025) use image-text soft labels and explicitly abandon the contrastive objective, arguing it causes catastrophic forgetting under small batches. **External-teacher distillation** such as KUEA (Gong et al., 2025) aligns CLIP with DINOv2 (Oquab et al., 2024) via kernel-based methods, but risks degrading the language-aligned contrastive structure. **Generative-model-based enhancement**—DIVA (Wang et al., 2025), un2CLIP (Li et al., 2025), GenHancer (Ma et al., 2025)—uses feedback from diffusion or unCLIP-style generative models to improve fine-grained visual perception, with large gains on MMVP. Their published results show that this need not come at the cost of CLIP’s other capabilities: DIVA and GenHancer report essentially unchanged zero-shot classification and retrieval, and un2CLIP reports improved text-to-image retrieval with reduced ImageNet accuracy (Appendix N). These methods rely on a large generative model during training and are therefore in a heavier compute class than single-epoch contrastive post-training. **Retrieval-augmented customization** such as REACT (Liu et al., 2023) retrieves task-relevant image-text pairs from a web-scale corpus to adapt CLIP to a target domain; it addresses domain customization rather than general-purpose refinement and is not directly comparable.

Our position. ComCLIP revisits the role of the contrastive loss within lightweight post-training at a fixed text encoder. It shows that the InfoNCE objective is a viable post-training signal when its temperature is set appropriately, and combines it with anchoring and relational distillation (Park et al., 2019; Tung and Mori, 2019; Zheng et al., 2021) from DINOv2. We do not aim to maximize fine-grained perception, where generative methods lead; our target is a single-epoch refinement that improves the transferability of the visual features while keeping the encoder a drop-in replacement.

3 Method

We propose **ComCLIP**, a lightweight post-training recipe that refines CLIP’s vision encoder under three losses that act on separable metric families (Sec. 1). The text encoder is frozen throughout to preserve drop-in compatibility with downstream VLMs. The framework is illustrated in Figure 2.

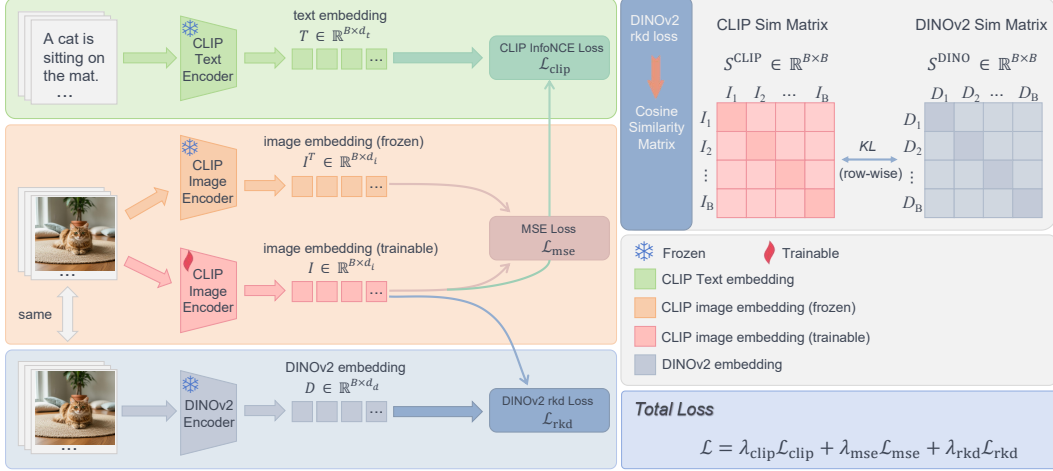


Figure 2: **Overview of ComCLIP.** Three parallel branches operate jointly: (i) frozen CLIP text encoder for image-text alignment ($\mathcal{L}_{\text{clip}}$), (ii) trainable CLIP image encoder anchored to its frozen pretrained counterpart ($\mathcal{L}_{\text{mse}}$), and (iii) frozen DINOv2 encoder providing relational supervision via batch-wise similarity matrices ($\mathcal{L}_{\text{rkd}}$). All three losses operate on the post-projection embedding.

3.1 Preliminaries

CLIP (Radford et al., 2021) consists of an image encoder f_v and a text encoder f_t , jointly trained on image-text pairs (x, t) via the symmetric InfoNCE loss (Oord et al., 2018):

$$\mathcal{L}_{\text{NCE}}(\mathbf{u}, \mathbf{v}; \tau) = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\mathbf{u}_i^\top \mathbf{v}_i / \tau)}{\sum_{j=1}^B \exp(\mathbf{u}_i^\top \mathbf{v}_j / \tau)}, \quad (1)$$

where $\mathbf{u}_i, \mathbf{v}_i \in \mathbb{R}^d$ are ℓ_2 -normalized projected embeddings, B is the batch size, and $\tau > 0$ is the contrastive temperature. The full CLIP objective is symmetric over images and texts. For each image x , we denote the post-projection ℓ_2 -normalized embedding produced by f_v as $\mathbf{u}(x) \in \mathbb{R}^d$, which lives in the shared image-text contrastive space. Given a pretrained CLIP (f_v^0, f_t^0) and a small post-training dataset $\mathcal{D}$, our goal is to refine f_v while keeping f_t^0 frozen, with all ComCLIP losses operating on $\mathbf{u}$.

3.2 Overview of ComCLIP

ComCLIP trains f_v under a weighted sum of three losses:

$$\mathcal{L}_{\text{ComCLIP}} = \lambda_{\text{clip}} \mathcal{L}_{\text{clip}} + \lambda_{\text{mse}} \mathcal{L}_{\text{mse}} + \lambda_{\text{rkd}} \mathcal{L}_{\text{rkd}}. \quad (2)$$

$\mathcal{L}_{\text{clip}}$ preserves *image-text alignment* via a properly-tempered contrastive objective; $\mathcal{L}_{\text{mse}}$ preserves the *pretrained feature manifold* by anchoring the post-projection embedding to its frozen pretrained counterpart; and $\mathcal{L}_{\text{rkd}}$ transfers the *inter-sample relational structure* of DINOv2 features via relational distillation. All three losses operate in the shared contrastive space, ensuring they jointly shape the same feature geometry. The design follows one principle: *refine relative structure* (which samples and texts are close to which) through $\mathcal{L}_{\text{clip}}$ and $\mathcal{L}_{\text{rkd}}$, while *anchoring absolute geometry* through $\mathcal{L}_{\text{mse}}$.

3.3 Properly-Tempered Contrastive Loss

We retain the standard contrastive objective (Eq. 1) as the primary post-training signal, but with a critical modification: we set the contrastive temperature τ to a value smaller than that of the CLIP pre-training initialization ($\tau \approx 0.07$, i.e., $1/\tau \approx 14$). Specifically, we fix $\tau = 0.01$ (i.e., $1/\tau = 100$) throughout post-training. The objective is:

$$\mathcal{L}_{\text{clip}} = \frac{1}{2} [\mathcal{L}_{\text{NCE}}(\mathbf{u}, \mathbf{v}; \tau) + \mathcal{L}_{\text{NCE}}(\mathbf{v}, \mathbf{u}; \tau)], \quad \tau = 0.01, \quad (3)$$

where $\mathbf{u}_i = f_v(x_i)$ and $\mathbf{v}_i = f_t^0(t_i)$.

Why a small temperature is necessary. Consider the image-to-text term for sample i , with similarities $s_{ij} = \mathbf{u}_i^\top \mathbf{v}_j$ and softmax weights $p_{ij} = \exp(s_{ij}/\tau) / \sum_k \exp(s_{ik}/\tau)$. Its gradient with respect to the logits is

$$\frac{\partial \mathcal{L}_i}{\partial s_{ij}} = \frac{1}{\tau} (p_{ij} - \mathbb{1}[j = i]), \quad p_{ij} \leq \exp\left(-\frac{s_{ii} - s_{ij}}{\tau}\right) \quad (j \neq i), \quad (4)$$

so the gradient on a negative pair is bounded by $\frac{1}{\tau} \exp(-\Delta_{ij}/\tau)$, where $\Delta_{ij} = s_{ii} - s_{ij}$ is its margin to the positive, and the gradient on the positive is $\frac{1}{\tau} \sum_{j \neq i} p_{ij}$. A pretrained CLIP already separates matched from mismatched pairs, i.e., $\Delta_{ij} > 0$ for most negatives in a post-training batch. The temperature then sets which pairs receive gradient, which is the hardness-aware property of the contrastive loss (Wang and Liu, 2021). With a small τ , Δ_{ij}/τ is large for all but the hardest negatives, so the gradient on easy pairs vanishes exponentially and the pretrained geometry around them is left intact (no forgetting), while the update concentrates on the few hard negatives whose margin is comparable to τ (refinement). With the pretraining initialization $\tau \approx 0.07$, typical margins are of the same order as τ , so a large fraction of already-correct pairs in every batch still receives gradient. The encoder is then pushed to re-fit the small post-training corpus rather than to correct its residual errors, which is the forgetting observed in prior work. This argument concerns which pairs are updated, not convergence. It is consistent with $\mathcal{L}_{\text{clip}}$ alone improving zero-shot accuracy from 61.82 to 62.90 at $\tau = 0.01$ and with every $1/\tau \geq 50$ improving over the base model (Sec. 4.5.3), and with a text-anchored geometric diagnostic in Appendix H. It is specific to the InfoNCE form of Eq. (1); objectives whose temperature plays a different role, such as the KL-based self-distillation of CLIP-Refine, are not covered by it.

Practical guidance. The argument gives a range rather than an optimal value: set τ well below the pretraining initialization, so that $\Delta_{ij}/\tau \gg 1$ for most in-batch negatives of the pretrained model. Empirically, every $50 \leq 1/\tau \leq 200$ improves zero-shot accuracy over the base model (Sec. 4.5.3), and $\tau = 0.01$ does so at every batch size from 128 to 2,048 (Appendix C); we transfer it unchanged to ViT-L/14, SigLIP, COCO Caption and CC12M. We do not derive an optimal τ for a given batch size or dataset.

3.4 MSE Anchoring Loss

To preserve the intra-modal visual feature manifold within the CLIP pretrained image-text contrastive space—which pure contrastive optimization on a small dataset tends to distort by overfitting to the post-training image-text correspondences—we introduce a feature-level MSE anchoring loss between the post-projection embeddings of the trainable encoder f_v and a frozen copy f_v^0 :

$$\mathcal{L}_{\text{mse}} = \frac{1}{B} \sum_{i=1}^B \|\mathbf{u}(x_i) - \mathbf{u}^0(x_i)\|_2^2. \quad (5)$$

By anchoring $\mathbf{u}$ to its pretrained counterpart $\mathbf{u}^0$, $\mathcal{L}_{\text{mse}}$ provides a stable reference in the contrastive space, preventing the encoder from drifting under the joint optimization of $\mathcal{L}_{\text{clip}}$ and $\mathcal{L}_{\text{rkd}}$.

3.5 Relational Distillation from DINOv2

To enrich the visual structure of CLIP’s features beyond what image-text pairs supervise, we distill from DINOv2 (Oquab et al., 2024), whose self-supervised features encode visual structure complementary to CLIP’s contrastive pretraining. Directly aligning CLIP’s features to DINOv2’s would force the CLIP feature space into DINOv2’s purely visual coordinate system—catastrophically degrading image-text alignment. Following relational knowledge distillation (Park et al., 2019; Tung and Mori, 2019), we instead distill the *relational structure* between samples within a batch.

Let $\mathbf{u}(x_i)$ and $\mathbf{g}(x_i)$ denote the ℓ_2 -normalized post-projection embeddings of the trainable CLIP encoder and the frozen DINOv2 encoder, respectively. We compute batch-wise self-similarity matrices and convert them into row-wise distributions via a temperature-scaled softmax:

$$S_{ij}^{\text{clip}} = \mathbf{u}(x_i)^\top \mathbf{u}(x_j), \quad S_{ij}^{\text{dino}} = \mathbf{g}(x_i)^\top \mathbf{g}(x_j), \quad (6)$$

$$P_{ij}^{\text{clip}} = \frac{\exp(S_{ij}^{\text{clip}}/T)}{\sum_k \exp(S_{ik}^{\text{clip}}/T)}, \quad P_{ij}^{\text{dino}} = \frac{\exp(S_{ij}^{\text{dino}}/T)}{\sum_k \exp(S_{ik}^{\text{dino}}/T)}. \quad (7)$$

The relational distillation loss is the row-wise KL divergence, the distributional relation-matching form used in ReSSL (Zheng et al., 2021):

$$\mathcal{L}_{\text{rkd}} = \frac{1}{B} \sum_{i=1}^B \text{KL}(P_{i,:}^{\text{dino}} \parallel P_{i,:}^{\text{clip}}), \quad (8)$$

with $T = 0.1$ and DINOv2 features computed without gradients. We compute $\mathcal{L}_{\text{rkd}}$ on the post-projection embedding so that all three losses shape the same space; the pre-projection alternative performs comparably (Appendix E). In a multi-seed control, adding $\mathcal{L}_{\text{rkd}}$ to $\mathcal{L}_{\text{clip}} + \mathcal{L}_{\text{mse}}$ significantly improves linear probing on ViT-L/14 (Sec. 4.5.2).

Relation to KUEA. KUEA (Gong et al., 2025) also distills DINOv2’s relational structure under an ℓ_2 anchor, but matches kernel values element-wise, $[k_1(\mathbf{u}_i, \mathbf{u}_j) - k_2(\mathbf{g}_i, \mathbf{g}_j)]^2$, with a trainable CLIP-side kernel to calibrate absolute similarity magnitudes. $\mathcal{L}_{\text{rkd}}$ is parameter-free and, as $\text{softmax}(\mathbf{z} + c) = \text{softmax}(\mathbf{z})$, constrains each sample’s relative neighborhood rather than absolute similarities. These are left to $\mathcal{L}_{\text{mse}}$, so the two terms do not compete for them, unlike value matching, which pulls them toward DINOv2’s scale. This is a design rationale; Table 4 compares full recipes, not a single-factor swap.

3.6 Complementary Effects of the Three Losses

$\mathcal{L}_{\text{clip}}$ and $\mathcal{L}_{\text{rkd}}$ are both row-wise softmax objectives over in-batch similarities, so both act on relative structure: with a small τ , $\mathcal{L}_{\text{clip}}$ refines how texts are ranked around each image (Sec. 3.3), and $\mathcal{L}_{\text{rkd}}$ adds DINOv2’s ranking of images around it, a self-normalizing target of the same kind. A kernel-value term such as KUEA’s instead fixes absolute magnitudes to DINOv2’s scale, against the relative structure that $\mathcal{L}_{\text{clip}}$ sharpens. $\mathcal{L}_{\text{mse}}$, the only term on absolute position, keeps both from over-shaping the features. Sec. 4.5.1 and Sec. 4.5.2 test this empirically.

4 Experiments

4.1 Settings

Pre-trained models. We experiment with two student backbones: OpenAI CLIP ViT-B/16 and ViT-L/14. DINOv2 ViT-L/14 (with registers) (Oquab et al., 2024) serves as the visual teacher for $\mathcal{L}_{\text{rkd}}$. During post-training, both the student’s text encoder and DINOv2 are kept frozen; only the student’s vision encoder is updated.

Post-training datasets. We use three datasets spanning different scales and curation styles: **CC3M** (Sharma et al., 2018) and **CC12M** (Changpinyo et al., 2021) (web-scraped, 3M and 12M pairs) and **COCO Caption** (Lin et al., 2014) (591K human-annotated pairs).

Baselines. We compare against (i) the original **OpenAI CLIP** without post-training, (ii) **CLIP-Refine** (Yamaguchi et al., 2025) trained on COCO Caption and on CC3M for 1 epoch, and (iii) **KUEA** (Gong et al., 2025) trained on ImageNet-1K (Deng et al., 2009) and on CC3M for 2 epochs (its default schedule), all reproduced from the official codebases with their original hyperparameters. Generative-model-based methods (DIVA, un2CLIP, GenHancer) are compared using their published numbers in Appendix N.

Training details. ComCLIP is implemented on top of the official KUEA codebase. We use AdamW (Loshchilov and Hutter, 2019) with learning rate 1×10^{-6} , weight decay 0.1, and a 410-step linear warm-up. Each model is post-trained for 1 epoch on 8 A800 GPUs with global batch size 1,024 and the standard per-sample CLIP data augmentation inherited from the KUEA codebase. Because augmentation is applied per sample, the features of the frozen CLIP copy and of DINOv2 are computed on the fly (forward pass only) rather than cached. The contrastive temperature is fixed at $\tau = 0.01$, and all three loss weights are set to $\lambda_{\text{clip}} = \lambda_{\text{mse}} = \lambda_{\text{rkd}} = 1.0$. The delivered model is a single CLIP vision encoder with the same architecture and parameter count as the input model, so inference cost is unchanged.

Evaluation. We evaluate on five task families: (i) zero-shot classification averaged over 12 datasets; (ii) image-text retrieval (R@1) on MS COCO (Chen et al., 2015) and Flickr30K (Young et al., 2014); (iii) linear probing averaged over 5 datasets; (iv) MMVP (Tong et al., 2024) for fine-grained visual

Table 1: Comparison of post-training methods on two backbones (ViT-B/16 and ViT-L/14). The “+” denotes the application of a post-training method and dataset to the pre-trained OpenAI CLIP. Each method is reported under its strongest validated setting (Sec. 4.2): ComCLIP and CLIP-Refine are trained for 1 epoch and KUEA for 2 epochs (its default schedule). Cells with a subscript are means $\pm$ standard deviation over seeds (3 seeds; see Table 2); all other cells are single runs. MMVP has $N = 135$ image pairs and a seed standard deviation of up to ~ 3 points, so single-run MMVP differences should not be over-interpreted. The best and second-best results within each backbone are in **bold** and underline; highlighting does not imply statistical significance.

Method	Zero-shot	MS COCO (R@1)		Flickr30K (R@1)		Linear Probe	MMVP
	CLS Avg	I→T	T→I	I→T	T→I	Avg	Avg
Backbone: ViT-B/16							
OpenAI CLIP (no post-training)	61.82	48.16	31.47	74.70	57.10	44.70	12.59
+ KUEA (ImageNet-1K)	62.07	48.48	31.99	75.00	57.62	44.65	11.85
+ KUEA (CC3M)	61.99	48.00	31.47	75.00	57.00	44.72	14.07
+ CLIP-Refine (COCO Caption)	62.87 ± 0.11	54.17 ± 0.28	37.72 ± 0.07	81.30 ± 0.40	64.45 ± 0.21	42.28 ± 0.62	16.05 ± 1.14
+ CLIP-Refine (CC3M)	61.59	41.12	33.29	66.20	60.18	44.89	19.26
+ ComCLIP (COCO Caption)	62.97	52.24	35.36	79.50	61.36	<u>47.23</u>	15.56
+ ComCLIP (CC3M)	<u>63.00</u> ± 0.13	<u>52.49</u> ± 0.28	<u>36.17</u> ± 0.02	77.80 ± 0.00	<u>63.11</u> ± 0.09	48.99 ± 0.35	14.81 ± 2.97
+ ComCLIP (CC12M)	63.04	51.58	35.56	77.10	62.42	46.56	15.56
Backbone: ViT-L/14							
OpenAI CLIP (no post-training)	66.12	50.04	34.29	77.90	59.80	54.90	17.78
+ KUEA (ImageNet-1K)	66.89	50.88	35.66	79.50	61.18	59.92	17.78
+ KUEA (CC3M)	66.36	50.30	34.68	78.00	60.02	55.82	18.52
+ CLIP-Refine (COCO Caption)	66.31	53.28	38.21	80.10	64.56	50.83	17.04
+ CLIP-Refine (CC3M)	—	—	—	—	—	—	19.01 ± 1.86
+ ComCLIP (COCO Caption)	67.09	<u>54.32</u>	38.79	<u>81.80</u>	64.98	58.28	20.74
+ ComCLIP (CC3M)	<u>67.75</u>	54.56	39.40	82.60	65.82	56.99	24.20 ± 0.86
+ ComCLIP (CC12M)	67.78	53.74	38.63	80.10	<u>65.00</u>	57.42	<u>22.22</u>

perception; and (v) downstream VLM evaluation on 8 benchmarks via LLaVA-1.5-7B integration. The full list of zero-shot and linear-probing datasets, along with the VLM benchmarks, is provided in Appendix A. MMVP contains only $N = 135$ image pairs, so one pair corresponds to 0.74 points and seed-to-seed variation is several points; we therefore base MMVP claims only on multi-seed comparisons (Sec. 4.3).

4.2 Main Results

Table 1 compares ComCLIP with the original OpenAI CLIP and two post-training baselines, KUEA (Gong et al., 2025) and CLIP-Refine (Yamaguchi et al., 2025), on ViT-B/16 and ViT-L/14. Since different post-training methods favor different datasets and training budgets, the table reports each method under its *strongest validated setting* rather than forcing a single shared configuration. Concretely, CLIP-Refine is trained on COCO Caption, where it performs best, and additionally on CC3M; KUEA is trained on ImageNet-1K and on CC3M for 2 epochs; ComCLIP and CLIP-Refine use 1 epoch, since 2 epochs gave no consistent improvement (Appendix G). As a result, the best-setting rows compare ComCLIP (CC3M) with CLIP-Refine (COCO Caption), i.e., methods trained on different data; the dataset-matched rows (both on CC3M for ViT-B/16, both on COCO Caption for ViT-L/14) are included in the same table and discussed below.

ComCLIP and CLIP-Refine occupy different points of a trade-off. On ViT-B/16, over three seeds, ComCLIP and CLIP-Refine are at parity on zero-shot classification (63.00 ± 0.13 vs. 62.87 ± 0.11 , $p = 0.26$), CLIP-Refine is ahead on all four retrieval metrics, and ComCLIP is far ahead on linear probing (48.99 ± 0.35 vs. 42.28 ± 0.62 , $p < 0.001$; Table 2). CLIP-Refine raises retrieval while reducing linear-probe accuracy below the original CLIP (44.70), whereas ComCLIP improves it. On ViT-L/14 the same pattern holds for linear probing (CLIP-Refine 50.83 vs. original 54.90; ComCLIP 56.99–58.28), and ComCLIP is also ahead on retrieval.

The linear-probe advantage is not an artifact of the training data. Under dataset-matched comparisons, ComCLIP retains a large linear-probe advantage over CLIP-Refine: with both methods trained on CC3M on ViT-B/16, 48.99 vs. 44.89 (+4.10), and with both trained on COCO Caption on ViT-L/14, 58.28 vs. 50.83 (+7.45).

KUEA yields small zero-shot changes (at most +0.77); its highest linear-probe accuracy (ViT-L/14, ImageNet-1K, 59.92) is plausibly helped by ImageNet-1K being in the probe suite, and on CC3M its gain is small (55.82).

Table 2: Multi-seed comparison between ComCLIP and CLIP-Refine (mean $\pm$ standard deviation). **Top:** ViT-B/16, 3 seeds per method, each method in its best setting (ComCLIP on CC3M, CLIP-Refine on COCO Caption). **Bottom:** ViT-L/14 MMVP, 3 seeds per method, both trained on CC3M. p : two-sided Welch t -test. MMVP has $N = 135$ image pairs; on ViT-B/16 the ComCLIP seeds correspond to $\{16, 20, 24\}$ correct pairs.

Method	Zero-shot	MS COCO (R@1)		Flickr30K (R@1)		Linear Probe	MMVP
	CLS Avg	I→T	T→I	I→T	T→I	Avg	Avg
ViT-B/16 (3 seeds)							
CLIP-Refine (COCO Caption)	62.87 $\pm$ 0.11	54.17 $\pm$ 0.28	37.72 $\pm$ 0.07	81.30 $\pm$ 0.40	64.45 $\pm$ 0.21	42.28 $\pm$ 0.62	16.05 $\pm$ 1.14
ComCLIP (CC3M)	63.00 $\pm$ 0.13	52.49 $\pm$ 0.28	36.17 $\pm$ 0.02	77.80 $\pm$ 0.00	63.11 $\pm$ 0.09	48.99 $\pm$ 0.35	14.81 $\pm$ 2.97
Welch p	0.26	2×10^{-3}	3×10^{-4}	4×10^{-3}	3×10^{-3}	4×10^{-4}	0.56
ViT-L/14 (3 seeds, both on CC3M)							
CLIP-Refine (CC3M)							19.01 $\pm$ 1.86
ComCLIP (CC3M)							24.20 $\pm$ 0.86
Welch p							0.025

Table 3: LLaVA-1.5-7B with the vision encoder swapped for ComCLIP ViT-L/14 (post-trained on CC3M), KUEA ViT-L/14 (ImageNet-1K), CLIP-Refine ViT-L/14 (COCO Caption), or the original OpenAI CLIP ViT-L/14; the LLM and projector are kept frozen. ‘*’ indicates official KUEA weights (Gong et al., 2025). Because the benchmarks differ in scale by more than $60\times$ (e.g., MME-p vs. RefCOCOg), we do not report a raw arithmetic mean. Instead, **Rel. Δ** is the mean per-benchmark relative change w.r.t. OpenAI CLIP (%), **Ceil. Δ** is the mean change after dividing each benchmark by its maximum attainable score (points), and **W/T/L** counts benchmarks improved/tied/degraded w.r.t. OpenAI CLIP. The best result in each benchmark column is in **bold**.

Method	Rel. Δ	Ceil. Δ	W/T/L	AI2D	POPE	RefCOCOg	V*	MME-c	MME-p	SQA-Img	OCRBench	TextVQA
OpenAI CLIP	–	–	–	52.75	85.20	20.67	41.36	320.00	1350.79	67.63	27.80	34.57
KUEA*	−1.23	−0.51	3/0/6	52.30	85.46	20.50	38.74	317.14	1350.82	66.68	27.40	34.69
CLIP-Refine	−0.95	−0.37	3/0/6	52.20	85.57	20.61	39.79	308.21	1345.97	67.72	27.60	34.84
ComCLIP	+0.02	+0.04	4/2/3	52.75	85.14	20.63	41.36	307.50	1373.47	67.82	28.20	34.89

Effect of the post-training corpus. Gains are not monotone in corpus size: on ViT-B/16, CC12M matches CC3M on zero-shot (63.04 vs. 63.00 $\pm$ 0.13) but is lower on linear probing (46.56 vs. 48.99 $\pm$ 0.35), and ViT-L/14 zero-shot is flat (67.75 vs. 67.78). We use CC3M by default (see Sec. 5).

4.3 Multi-seed Statistics

Table 2 reports seed statistics for the comparisons the paper relies on. On ViT-B/16 (three seeds), the only significant difference between ComCLIP and CLIP-Refine in favor of ComCLIP is linear probing (Welch $t = 16.3$, $p < 0.001$); CLIP-Refine is significantly ahead on all four retrieval metrics ($p < 0.005$); the zero-shot difference is not significant ($p = 0.26$), and on MMVP the two are indistinguishable (14.81 $\pm$ 2.97 vs. 16.05 $\pm$ 1.14, $p = 0.56$). On ViT-B/16, the ComCLIP MMVP seeds span [11.85, 17.78], and every MMVP value in our ViT-B/16 ablations (Sec. 4.5) lies within this range; we therefore draw no conclusions from MMVP on ViT-B/16. On ViT-L/14, with three seeds per method and both methods trained on CC3M, ComCLIP improves MMVP over CLIP-Refine (24.20 $\pm$ 0.86 vs. 19.01 $\pm$ 1.86, Welch $t = 4.4$, $p = 0.025$); this is the only MMVP claim we make. We note that generative enhancement methods report higher MMVP on the same backbone (Appendix N).

4.4 Downstream VLM Compatibility

To verify that the refined encoder remains a drop-in replacement, we replace LLaVA-1.5-7B’s vision encoder (originally CLIP ViT-L/14-336) with ComCLIP, KUEA, CLIP-Refine and the original OpenAI CLIP (all ViT-L/14-224), while keeping the LLM and cross-modal projector strictly frozen (Table 3). Because the benchmarks differ in scale by more than an order of magnitude, we aggregate with scale-normalized measures rather than a raw arithmetic mean. Under these measures, ComCLIP produces *no net change* relative to the original CLIP: the mean per-benchmark relative change is +0.02%, the ceiling-normalized change is +0.04 points, and it improves 4, ties 2, and degrades 3 of the 9 scores; excluding the two MME scores, the mean change is +0.12 points (+0.34%). The largest individual movements are on MME, with perception up by 22.68 and cognition down by 12.50, i.e., by more than twice as much in relative terms (+1.7% vs. −3.9%). By contrast, KUEA

Table 4: Loss decoupling ablation on ViT-B/16, post-trained on CC3M with batch size 128×8 . All configurations of $\{\mathcal{L}_{\text{clip}}, \mathcal{L}_{\text{mse}}, \mathcal{L}_{\text{rkd}}\}$ are trained for 1 epoch. KUEA and KUEA + $\mathcal{L}_{\text{clip}}$ use KUEA’s default 2-epoch schedule and hyperparameters, which were not re-tuned (loss weight, learning rate, anchor weight) after adding $\mathcal{L}_{\text{clip}}$. Temperature is fixed at $\tau = 0.01$ throughout. Entries are single runs except where a seed standard deviation is given. Best results per column (except MMVP) in **bold**. All MMVP values ($N = 135$) lie within the 3-seed range of ComCLIP ($[11.85, 17.78]$), so MMVP is not highlighted and no conclusion is drawn from it.

Method	Zero-shot 12 datasets Avg	MS COCO (R@1)		Flickr30K (R@1)		Linear Probe 5 datasets Avg	MMVP Avg
		I→T	T→I	I→T	T→I		
OpenAI CLIP (no post-training)	61.82	48.16	31.47	74.70	57.10	44.70	12.59
+ KUEA	61.99	48.00	31.47	75.00	57.00	44.72	14.07
+ (KUEA + $\mathcal{L}_{\text{clip}}$)	62.24	48.84	34.20	75.30	60.80	45.25	13.33
+ $\mathcal{L}_{\text{clip}}$	62.90	51.92	38.28	76.80	64.56	44.04	14.07
+ $\mathcal{L}_{\text{mse}}$	61.82	48.18	31.45	74.70	57.12	44.79	11.85
+ $\mathcal{L}_{\text{rkd}}$	56.64	43.30	25.57	69.30	49.66	44.42	17.78
+ ($\mathcal{L}_{\text{mse}} + \mathcal{L}_{\text{rkd}}$)	61.36	49.98	31.53	75.40	58.08	45.50	15.56
+ ($\mathcal{L}_{\text{clip}} + \mathcal{L}_{\text{mse}}$)	63.35	51.64	37.03	77.00	64.34	48.60	14.07
+ ($\mathcal{L}_{\text{clip}} + \mathcal{L}_{\text{rkd}}$)	62.49	51.50	36.67	77.30	63.96	47.15	14.07
+ ComCLIP ($\mathcal{L}_{\text{clip}} + \mathcal{L}_{\text{mse}} + \mathcal{L}_{\text{rkd}}$)	63.15	52.20	36.18	77.80	63.22	48.74	14.81 $\pm$ 2.97

and CLIP-Refine reduce the aggregate (-1.23% and -0.95% relative, each degrading 6 of 9 scores). We read these results as evidence of compatibility: ComCLIP’s refinement of the vision encoder does not break the frozen projector and LLM trained on the original CLIP, but it does not by itself improve downstream VLM accuracy without re-alignment.

4.5 Ablation Study

Ablations are run on ViT-B/16 with CC3M. Unless a seed standard deviation is given, entries are single runs; for reference, the ViT-B/16 seed standard deviations of ComCLIP are 0.13 (zero-shot), 0.35 (linear probe) and 2.97 (MMVP). Every ViT-B/16 MMVP value in this section lies within the seed range of a single configuration, so we report MMVP for completeness only.

4.5.1 Loss Decoupling

Table 4 reports all single-loss, pairwise, and full configurations of $\{\mathcal{L}_{\text{clip}}, \mathcal{L}_{\text{mse}}, \mathcal{L}_{\text{rkd}}\}$ (Finding 2). **Each loss moves a different group of metrics.** $\mathcal{L}_{\text{clip}}$ alone improves zero-shot classification (+1.08) and retrieval but leaves linear probing essentially unchanged (44.04 vs. 44.70). $\mathcal{L}_{\text{mse}}$ alone is a no-op (differences within 0.1). $\mathcal{L}_{\text{rkd}}$ alone severely degrades zero-shot classification (-5.18) and retrieval (-7.44 on Flickr30K T→I): distilling DINOv2’s relational structure without an image-text signal pulls the embedding away from the text space. **Combinations.** $\mathcal{L}_{\text{clip}} + \mathcal{L}_{\text{mse}}$ achieves the highest zero-shot accuracy (63.35) and a large linear-probe gain over $\mathcal{L}_{\text{clip}}$ alone (48.60 vs. 44.04): anchoring the absolute geometry lets the contrastive signal improve alignment without eroding transferable features. $\mathcal{L}_{\text{clip}} + \mathcal{L}_{\text{rkd}}$ improves linear probing less (47.15), and $\mathcal{L}_{\text{mse}} + \mathcal{L}_{\text{rkd}}$ leaves zero-shot accuracy below the baseline. The full objective reaches the highest linear-probe accuracy (48.74) with zero-shot accuracy (63.15) within ~ 1.5 seed standard deviations of the best variant. Adding properly-tempered $\mathcal{L}_{\text{clip}}$ to KUEA, without re-tuning KUEA’s hyperparameters, also improves its zero-shot accuracy (+0.25) and retrieval (up to +3.80 on Flickr30K T→I), so the temperature finding is not tied to ComCLIP’s other components; ComCLIP remains ahead of KUEA + $\mathcal{L}_{\text{clip}}$ on zero-shot (+0.91) and linear probing (+3.49), a comparison between recipes rather than a single-factor ablation (see the caption of Table 4).

4.5.2 Contribution of $\mathcal{L}_{\text{rkd}}$

Because the single-run effect of $\mathcal{L}_{\text{rkd}}$ above is small, we isolate it with a 3-seed control that adds $\mathcal{L}_{\text{rkd}}$ to $\mathcal{L}_{\text{clip}} + \mathcal{L}_{\text{mse}}$ (Appendix D, Table 9). On ViT-L/14, $\mathcal{L}_{\text{rkd}}$ significantly improves linear probing ($56.36 \pm 0.23 \rightarrow 57.12 \pm 0.14$, Welch $t = 4.9$, $p \approx 0.013$); on ViT-B/16 the effect has the same sign but is not significant ($48.54 \pm 0.10 \rightarrow 48.94 \pm 0.41$, $p \approx 0.23$), consistent with a benefit that grows with student capacity. It is not an all-metric gain: on ViT-L/14, zero-shot accuracy is unchanged ($67.76 \rightarrow 67.75$) and retrieval changes are small and mixed (e.g., COCO T→I $40.37 \rightarrow 39.40$). We

Table 5: Ablation study on the initialization of the temperature parameter (τ) in the CLIP loss. We conduct this experiment **exclusively using the CLIP loss** on the pre-trained OpenAI CLIP ViT-B/16 model to isolate the impact of τ . We conduct post-training on the CC3M dataset. The global batch size is fixed at 128×8 (1024) across all settings. The best results are highlighted in **bold**. MMVP ($N = 135$, single runs) is reported for completeness only: its variation across rows is within seed-to-seed variance (Table 2), so it is not highlighted and not used to select τ .

τ	Zero-shot 12 datasets Avg	MS COCO (R@1)		Flickr30K (R@1)		Linear Probe 5 datasets Avg	MMVP Avg
		I→T	T→I	I→T	T→I		
OpenAI CLIP (no post-training)	61.82	48.16	31.47	74.70	57.10	44.70	12.59
0.07	57.90	39.10	34.12	65.70	58.86	43.95	16.30
0.03	61.51	48.94	37.01	74.50	63.54	43.12	13.33
0.02	62.57	51.54	38.33	76.70	65.00	43.98	13.33
0.01	62.90	52.50	38.26	76.80	64.56	44.04	14.07
0.007	62.72	50.72	37.74	75.90	64.28	44.53	17.78
0.005	62.36	49.38	37.48	75.10	64.10	45.90	14.81

therefore retain $\mathcal{L}_{\text{rkd}}$ on the basis of linear-probe transferability only. The benefit also holds with a DINOv2-Base teacher, but our comparison does not single out a teacher size (Appendix D).

4.5.3 Effect of Contrastive Temperature

To examine Finding 1, we train with $\mathcal{L}_{\text{clip}}$ only and sweep τ (Table 5). **The pretraining temperature reproduces catastrophic forgetting.** At $\tau = 0.07$ (the CLIP pretraining initialization), zero-shot accuracy drops by 3.92 and Flickr30K I→T by 9.00, consistent with the gradient analysis in Sec. 3.3. **Sufficiently small τ enables successful post-training.** For every tested $50 \leq 1/\tau \leq 200$, contrastive post-training *improves* zero-shot accuracy (all in [62.36, 62.90]) and retrieval: the *magnitude* of τ , not its exact value, matters. We adopt $\tau = 0.01$ (best zero-shot, best or near-best retrieval; MMVP not used). **Batch size and learnability.** With $\tau = 0.01$, larger batches still help (+1.03 zero-shot from 128 to 2,048), but far less than the −3.92 caused by $\tau = 0.07$, so limited negatives alone do not explain the failures reported by CLIP-Refine (Yamaguchi et al., 2025); fixed and learnable τ differ by at most 0.05 (Appendix C).

5 Conclusion and Limitations

We presented **ComCLIP**, a lightweight single-epoch CLIP post-training recipe at a fixed text encoder. We showed that, for InfoNCE, the forgetting attributed to contrastive post-training is driven primarily by the temperature magnitude rather than by small batches: with a small τ , already-separated pairs receive vanishing gradient and contrastive post-training improves the pretrained model. Relative to CLIP-Refine, ComCLIP is at parity on zero-shot classification, significantly better on linear-probe transferability (on both backbones and under dataset-matched data), better on MMVP on ViT-L/14, and behind on retrieval; inside LLaVA-1.5-7B it is a drop-in replacement with no net change.

Limitations and Future Work. (i) ComCLIP targets CLIP-like models with a shared contrastive image-text space; extending it to vision encoders or VLMs without such a space is future work. The temperature analysis covers InfoNCE only; it does not directly apply to the KL-based self-distillation of CLIP-Refine or to SigLIP’s sigmoid loss (Appendix B applies our InfoNCE objective to a SigLIP backbone). (ii) MMVP has only 135 pairs and large seed variance; our MMVP gain over CLIP-Refine is significant only on ViT-L/14, and generative methods report substantially higher MMVP (Appendix N). ComCLIP is not a method for maximizing fine-grained perception. (iii) Our downstream evaluation uses a single VLM configuration (LLaVA-1.5-7B, frozen projector and LLM); other VLM architectures would better establish generality. Without re-aligning the projector, ComCLIP preserves but does not improve downstream VLM accuracy; whether re-alignment converts its linear-probe gains into VLM gains is untested. (iv) Performance is not monotone in corpus size (Sec. 4.2); whether this reflects single-epoch dynamics, data quality, or a capacity limit is open. (v) Training runs a frozen CLIP copy and a DINOv2 teacher forward at every step, and we report no per-component compute breakdown; the efficiency we claim is a single epoch on a small corpus with unchanged inference cost.

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A Experiments Settings

Datasets Settings For zero-shot evaluation, we use the following datasets: ImageNet-1K, CIFAR-10/100 (Krizhevsky et al., 2009), Caltech101 (Fei-Fei et al., 2004), FER2013 (Goodfellow et al., 2013), OxfordPets (Parkhi et al., 2012), DTD (Cimpoi et al., 2014), RESISC45 (Cheng et al., 2017), EuroSAT (Helber et al., 2019), PCAM (Veeling et al., 2018), ImageNet-Sketch (Wang et al., 2019), ImageNet-O (Hendrycks et al., 2021). We evaluate the model’s linear probing performance on five datasets: ImageNet-1K, SVHN Netzer et al. (2011), GTSRB Stallkamp et al. (2011), CLEVR Distance, and CLEVR Counts Johnson et al. (2017). For multimodal large language model evaluation, we select the following benchmarks: AI2D Kembhavi et al. (2016), POPE Li et al. (2023b), RefCOCOg Mao et al. (2016), VSTAR Wu and Xie (2024), MME Fu et al. (2023), SQA-Img Lu et al. (2022), OCRBench Liu et al. (2024b), and TextVQA Singh et al. (2019).

B Post-training Performance Results based on SigLIP

Table 6: Performance evaluation of post-training methods based on the SigLIP backbone. We compare our proposed method (**ComCLIP**) against the pre-trained SigLIP ViT-SO400M/14 baseline and KUEA (post-trained on ImageNet-1K). All methods are evaluated on the CC3M dataset for post-training, except for the KUEA baseline which is evaluated on its original setting. ‘*’ denotes results reported by the original authors(Gong et al., 2025). The best results in each column are highlighted in **bold**.

Method	Zero-shot CLS Avg	MS COCO (R@1) I→T	MS COCO (R@1) T→I	Flickr30K (R@1) I→T	Flickr30K (R@1) T→I	Linear Probe Avg	MMVP Avg
SigLIP (no post-training)	71.24	68.16	52.20	88.90	74.80	70.87	39.26
KUEA (ImageNet-1K)*	71.67	-	-	-	-	-	-
ComCLIP (CC3M)	72.15	70.06	53.05	89.50	75.78	70.88	41.48

Applying ComCLIP to a SigLIP backbone.

We apply ComCLIP to the pre-trained SigLIP ViT-SO400M/14, post-training on CC3M for one epoch with τ fixed at 0.01 (Table 6, single run). Note that this experiment uses our InfoNCE contrastive objective on a SigLIP backbone; it does not test whether the temperature finding transfers to SigLIP’s sigmoid loss, which we leave to future work. Compared with the pre-trained SigLIP and with KUEA (zero-shot result reported in the original paper), ComCLIP improves zero-shot classification (71.24 $\rightarrow$ 72.15) and retrieval, and leaves linear probing unchanged (70.87 $\rightarrow$ 70.88); the MMVP change (39.26 $\rightarrow$ 41.48, three pairs) is within the seed variance we observe elsewhere. Figure 3 visualizes these results.

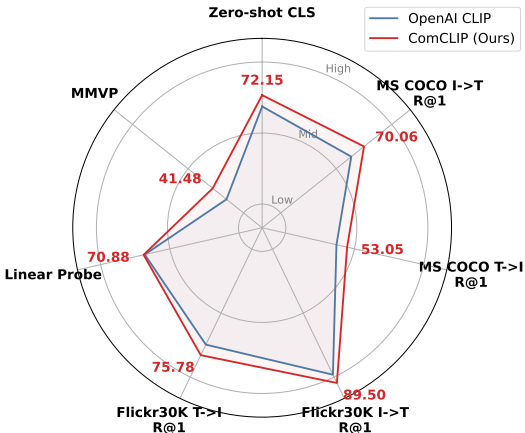


Figure 3: Multi-dimensional performance comparison on SigLIP ViT-SO400M/14.

C Disentangling Temperature from Batch Size and Learnability

To rule out two alternative explanations of Finding 1—that the underlying issue lies with batch size or with the learnability of τ —we run two controlled studies using only $\mathcal{L}_{\text{clip}}$ with $\tau = 0.01$ on ViT-B/16.

Batch size. CLIP-Refine (Yamaguchi et al., 2025) attributes the failure of contrastive post-training mainly to insufficient negatives, reporting that even batch size 2,048 cannot recover CLIP’s performance. With $\tau = 0.01$, contrastive post-training improves zero-shot accuracy over the original CLIP at every tested batch size from 128 to 2,048 (Table 7). Larger batches still provide gains (+1.03 zero-shot from 128 to 2,048), but these are much smaller than the degradation caused by an improperly set

Table 7: Ablation study on the effect of batch size during post-training with the CLIP loss. Experiments are conducted on the pre-trained OpenAI CLIP ViT-B/16 model, utilizing the CC3M dataset exclusively for post-training. Only the CLIP loss is applied to update the visual encoder ($\lambda_{\text{clip}} = 1.0, \lambda_{\text{mse}} = 0.0, \lambda_{\text{rkd}} = 0.0$). The batch sizes are denoted as (*per-GPU batch size* $\times$ *number of GPUs*). The best results in each column are highlighted in **bold**. MMVP ($N = 135$, single runs) varies within seed-to-seed variance (Table 2) and is not highlighted.

Batch Size	Zero-shot 12 datasets Avg	MS COCO (R@1)		Flickr30K (R@1)		Linear Probe 5 datasets Avg	MMVP Avg
		I→T	T→I	I→T	T→I		
16 $\times$ 8	62.00	50.56	37.26	76.10	63.22	42.15	15.56
32 $\times$ 8	62.50	51.26	37.79	76.50	64.16	42.98	11.85
64 $\times$ 8	62.65	51.44	38.10	76.50	64.12	43.65	14.81
128 $\times$ 8	62.69	52.50	38.26	77.40	64.52	43.68	15.56
256 $\times$ 8	63.03	52.52	38.22	77.80	64.80	44.63	18.52

Table 8: Ablation study on the learnability of τ and the impact of different post-training datasets (COCO Caption vs. CC3M). All experiments are conducted on the pre-trained OpenAI CLIP ViT-B/16 model, utilizing exclusively the CLIP loss ($\lambda_{\text{clip}} = 1.0, \lambda_{\text{mse}} = 0.0, \lambda_{\text{rkd}} = 0.0$). During post-training, only the parameters of the visual encoder are updated while the text encoder remains strictly frozen. The best results in each metric are highlighted in **bold** (MMVP, single runs within seed-to-seed variance, is not highlighted).

Dataset	Learnable	Batch Size	Zero-shot 12 datasets Avg	MS COCO (R@1)		Flickr30K (R@1)		Linear Probe 5 datasets Avg	MMVP Avg
				I→T	T→I	I→T	T→I		
COCO Caption	Yes	128 $\times$ 8	62.25	49.86	34.74	77.20	61.38	45.30	14.81
	No	128 $\times$ 8	62.25	49.86	34.74	77.20	61.40	45.27	14.81
CC3M	Yes	128 $\times$ 8	62.95	52.22	38.22	76.90	64.48	43.42	14.07
	No	128 $\times$ 8	62.90	52.50	38.26	76.80	64.56	44.04	14.07

Table 9: Contribution of $\mathcal{L}_{\text{rkd}}$ and choice of teacher. **Top:** adding $\mathcal{L}_{\text{rkd}}$ (DINOv2-Large teacher) to $\mathcal{L}_{\text{clip}} + \mathcal{L}_{\text{mse}}$, linear probe (5-dataset average), mean $\pm$ standard deviation over 3 seeds, CC3M, 1 epoch; p : two-sided Welch t -test. **Bottom:** teacher choice for $\mathcal{L}_{\text{rkd}}$ (single runs, CC3M, 1 epoch). The ViT-B/16 teacher runs use the same configuration as Table 4.

Linear probe with vs. without $\mathcal{L}_{\text{rkd}}$ (3 seeds)			
Backbone	$\mathcal{L}_{\text{clip}} + \mathcal{L}_{\text{mse}}$	+ $\mathcal{L}_{\text{rkd}}$ (ComCLIP)	Welch p
ViT-B/16	48.54 $\pm$ 0.10	48.94 $\pm$ 0.41	0.23
ViT-L/14	56.36 $\pm$ 0.23	57.12 $\pm$ 0.14	0.013
Teacher for $\mathcal{L}_{\text{rkd}}$ (single runs)			
Configuration	Zero-shot (12 avg)	Linear Probe (5 avg)	
ViT-B/16, $\mathcal{L}_{\text{clip}} + \mathcal{L}_{\text{mse}}$ (no teacher)	63.35	48.60	
ViT-B/16, + $\mathcal{L}_{\text{rkd}}$, MAE-Large	63.51	46.99	
ViT-B/16, + $\mathcal{L}_{\text{rkd}}$, DINOv2-Base	62.87	49.28	
ViT-B/16, + $\mathcal{L}_{\text{rkd}}$, DINOv2-Large	63.15	48.74	
ViT-L/14, + $\mathcal{L}_{\text{rkd}}$, DINOv2-Base	—	56.91	
ViT-L/14, + $\mathcal{L}_{\text{rkd}}$, DINOv2-Large	67.75	56.99	

τ (−3.92). Temperature calibration is therefore a critical and previously under-emphasized factor in InfoNCE post-training, and limited negatives alone do not explain the reported failures.

Learnability. On COCO Caption and CC3M, with τ initialized at 0.01 (Table 8), fixed and learnable τ yield near-identical results (within 0.05 on zero-shot): initialized at a sufficiently small value, τ does not drift back towards the pretraining value. The critical factor is the magnitude of τ , not whether it remains learnable.

D Contribution of $\mathcal{L}_{\text{rkd}}$ and Choice of Teacher

Table 9 (top) gives the 3-seed control discussed in Sec. 4.5.2. The teacher comparison (bottom) does not single out a teacher. On ViT-B/16, DINOv2-Base gives the highest linear-probe accuracy (49.28)

and MAE-Large the highest zero-shot accuracy (63.51) in single runs; on ViT-L/14, DINOv2-Base and DINOv2-Large give nearly identical linear probing (56.91 vs. 56.99). The benefit of $\mathcal{L}_{\text{rkd}}$ is thus present even when teacher and student have matched capacity. We use DINOv2-Large for both backbones for uniformity. These results support the component but do not by themselves select a teacher size, and a cheaper DINOv2-Base teacher is a reasonable alternative.

E Where to Apply RKD

Table 10: Investigation of the alignment position for DINOv2 rkd loss features on ViT-B/16. We conduct post-training on the CC3M dataset. ComCLIP-bp denotes that the visual features of the student model used for rkd loss calculation are extracted prior to the projection layer. ComCLIP represents our default configuration, where visual features are projected into the joint image-text embedding space via the original projection layer. Single runs except where a seed standard deviation is given; the MMVP difference is within seed-to-seed variance (Table 2) and is not highlighted.

Method	Zero-shot 12 datasets Avg	MS COCO (R@1)		Flickr30K (R@1)		Linear Probe 5 datasets Avg	MMVP Avg
		I→T	T→I	I→T	T→I		
OpenAI CLIP (no post-training)	61.82	48.16	31.47	74.70	57.10	44.70	12.59
ComCLIP-bp	63.19	51.44	36.54	77.00	63.50	48.23	13.33
ComCLIP	63.15	52.20	36.18	77.80	63.22	48.74	14.81 $\pm$ 2.97

We additionally investigate where DINOv2 distillation should be applied along the visual encoder. ComCLIP’s default configuration applies $\mathcal{L}_{\text{rkd}}$ on the post-projection embedding $u(x_i)$ (the same space used by $\mathcal{L}_{\text{clip}}$). A natural alternative, denoted **ComCLIP-bp**, applies it on the pre-projection [CLS] feature $h(x_i)$, which retains the full pretrained visual semantics before dimensionality reduction.

Table 10 compares the two choices on ViT-B/16 (CC3M, 1 epoch). The two variants perform comparably: differences on zero-shot classification, retrieval and linear probing are within ~ 0.8 points, and the MMVP difference is within seed-to-seed variance (Table 2), so we draw no conclusion from it. We adopt the post-projection variant as the default so that all three losses act on the same embedding space.

F Hyperparameter Selection Protocol

Table 11: Ablation study on the balancing coefficients of the three loss components used during post-training. Setting the weight of the primary CLIP loss as the baseline ($\lambda_{\text{clip}} = 1.0$), we investigate the impact of varying the proportions of the MSE loss (λ_{mse}) and RKD loss (λ_{rkd}) coefficients. All experiments are conducted on the CC3M dataset based on the pre-trained OpenAI CLIP ViT-B/16 architecture. The best results in each column are highlighted in **bold**; MMVP (single runs except where a seed standard deviation is given) is within seed-to-seed variance and is not highlighted.

Loss Weights			Zero-shot 12 datasets Avg	MS COCO (R@1)		Flickr30K (R@1)		Linear Probe 5 datasets Avg	MMVP Avg
λ_{clip}	λ_{mse}	λ_{rkd}		I→T	T→I	I→T	T→I		
1.0	1.0	0.5	62.77	52.42	36.47	78.00	63.06	49.75	14.81
1.0	0.5	1.0	62.91	52.40	36.56	77.50	63.32	47.29	13.33
1.0	0.5	0.5	63.03	52.62	36.76	77.80	63.38	48.85	17.04
1.0	1.0	1.0	63.15	52.20	36.18	77.80	63.22	48.74	14.81 $\pm$ 2.97

To ensure no validation leakage, we determine all hyperparameters on the ViT-B/16 + CC3M setting only, using the following protocol: **Temperature** τ : Selected via the sweep in Table 5. We chose $\tau = 0.01$ as it gives the best zero-shot accuracy and the best or near-best retrieval; MMVP was not used for selection. We then *transferred this value unchanged* to ViT-L/14, COCO Caption, and CC12M experiments, without any additional tuning.

We perform an ablation study to investigate the influence of the weighting coefficients (λ_{clip} , λ_{mse} , λ_{rkd}) for the three loss components. Taking the CLIP loss as our primary objective ($\lambda_{\text{clip}} = 1.0$), we systematically vary the scales of λ_{mse} and λ_{rkd} to observe their collective impact on model performance. The results are summarized in Table 11.

Different configurations favor different metrics: $\lambda_{\text{clip}} : \lambda_{\text{mse}} : \lambda_{\text{rkd}} = 1.0 : 1.0 : 0.5$ gives the best linear-probe accuracy, and $1.0 : 1.0 : 1.0$ the best zero-shot accuracy. The variation across ratios is small: within 0.4 points on zero-shot classification and within 0.6 points on retrieval. We adopt $1.0 : 1.0 : 1.0$ as the default for simplicity; MMVP was not used for this choice.

G Results After 1 or 2 epochs of Post-training

Table 12: Performance evaluation of post-training with ComCLIP on the CC3M dataset. Experiments are based on the pre-trained OpenAI CLIP ViT-B/16 architecture. We compare the results obtained after 1 and 2 epochs of post-training against the OpenAI CLIP baseline. The best results in each column are highlighted in **bold**; MMVP (single run for 2 epochs, 3-seed mean for 1 epoch) is within seed-to-seed variance and is not highlighted.

Method	epochs	Zero-shot CLS Avg	MS COCO (R@1)		Flickr30K (R@1)		Linear Probe Avg	MMVP Avg
			I→T	T→I	I→T	T→I		
OpenAI CLIP (Baseline)	-	61.82	48.16	31.47	74.70	57.10	44.70	12.59
ComCLIP	1	63.15	52.20	36.18	77.80	63.22	48.74	14.81 $\pm$ 2.97
ComCLIP	2	62.84	52.02	36.24	77.20	63.44	49.98	16.30

To investigate the impact of multiple epochs, we train ComCLIP for 1–2 epochs on the ViT-B/16 model using the CC3M dataset. Table 12 indicates that extending the training duration did not yield a significant improvement in overall performance, so we use a single epoch throughout.

H Geometric Metric for Preserving the Pre-trained Image-Text Feature Space

We treat the frozen text features as anchor points and compute the KL divergence of the image feature distribution before and after post-training—referred to as Text-Anchored KL Divergence. A lower KL divergence indicates that the manifold of image-text feature in contrastive space has been better preserved.

Table 13 demonstrates that when utilizing the temperature ($\tau = 0.07$) used for initialization during the CLIP pre-training phase, the KL divergence reaches a value of 5.50; this indicates that, relative to the frozen text anchors, the visual features have undergone severe manifold drift. Conversely, the "small τ " mechanism we adopted effectively suppresses this drift, resulting in a KL divergence of only 0.48. This is consistent with the gradient analysis in Sec. 3.3: with a small τ , already-separated pairs receive vanishing gradient, so the image features move little relative to the frozen text anchors. The KL is not monotone in τ (it rises again at $\tau = 0.005$), matching the slight decline in zero-shot accuracy at the smallest temperature.

I Training Curves of $\mathcal{L}_{\text{clip}}$ under Different τ Settings

Figure 4 visualizes the variation of $\mathcal{L}_{\text{clip}}$ as a function of training steps under different settings of τ . When using the temperature $\tau = 0.07$ (i.e., $\text{logit_scale} \approx 14$)—the value initialized by the pre-trained CLIP model— $\mathcal{L}_{\text{clip}}$ remains relatively high and fails to converge to a lower value within 3,000 steps. As τ is decreased (equivalently, as logit_scale is increased to 33, 50, and 100), $\mathcal{L}_{\text{clip}}$ gradually declines; notably, the convergence curve for $\text{logit_scale} = 100$ is the most optimal. When τ is further reduced—corresponding to an increase in logit_scale to 142 and 200— $\mathcal{L}_{\text{clip}}$ increases slightly; nevertheless, it still converges to a favorable level within 3,000 steps, demonstrating that an appropriately small value of τ is effective for the post-training process.

J Detailed Zero-shot Experimental Results

Table 14 presents per-dataset single-run results for the baselines and ComCLIP across the 12 zero-shot classification benchmarks. On ViT-B/16, ComCLIP and CLIP-Refine (COCO Caption) are at parity in average zero-shot accuracy; over three seeds the difference (63.00 $\pm$ 0.13 vs. 62.87 $\pm$ 0.11) is not significant (Table 2). When trained on CC3M, CLIP-Refine drops to 61.59, notably on ImageNet-1K

Table 13: Text-anchored geometric diagnostic of models post-trained using only $\mathcal{L}_{\text{clip}}$ across different temperature (τ) values. A lower Text Anchor KL ($\downarrow$, closer to 0) indicates that the post-trained visual feature manifold has not suffered from severe structural drift relative to the frozen text features.

τ	$\text{logit Scale } (1/\tau)$	Text Anchor KL ($\downarrow$)
0.07	14.29	5.50
0.03	33.33	2.18
0.02	50.00	1.25
0.01	100.00	0.48
0.007	142.86	0.56
0.005	200.00	1.15

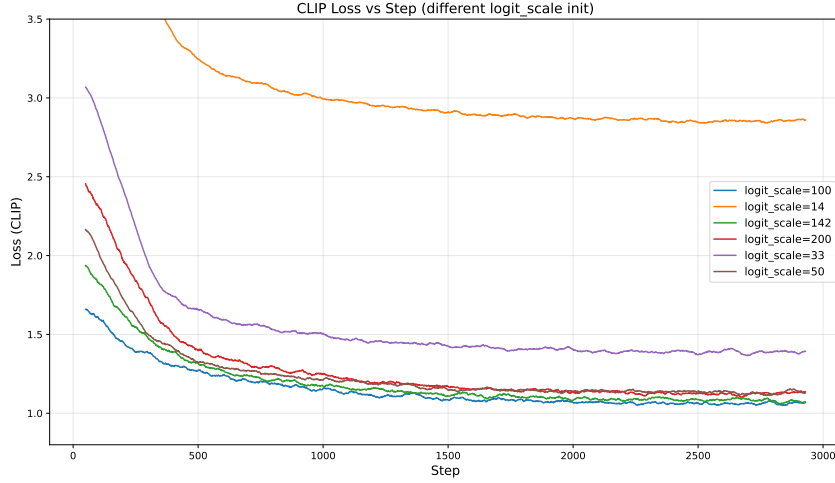


Figure 4: $\mathcal{L}_{\text{clip}}$ **Training Curve**. With the temperature τ ($\text{logit_scale} = 1/\tau$) set to various values, post-training was performed using only $\mathcal{L}_{\text{clip}}$. Convergence of the loss function was optimal when $\text{logit_scale} = 100$.

Table 14: Top-1 zero-shot classification accuracy on 12 public benchmarks. The best and second-best results in each model block are highlighted in **bold** and underline, respectively.

Model	Dataset	epochs	Method	Avg	IN-1k	Cifar10	Cifar100	CalTech	FER	Pets	DTD	RESISC	EuroSAT	PCAM	IN-S	IN-O
ViT-B/16	—	-	OpenAI CLIP	61.82	68.35	90.00	65.61	82.17	46.39	88.99	44.95	58.19	55.91	50.73	48.24	42.30
	ImageNet-1k	2	KUEA	62.07	68.52	90.69	67.08	82.14	46.45	89.21	45.00	58.14	55.26	51.01	48.23	43.15
	CC3M	2	KUEA	61.99	68.40	90.27	66.52	82.19	46.53	89.02	45.05	58.33	56.00	51.12	48.25	42.15
	COCO Caption	1	CLIP-Refine	62.99	67.74	90.50	68.28	83.40	49.80	89.75	44.68	59.35	56.20	55.18	48.14	42.90
	CC3M	1	CLIP-Refine	61.59	63.30	89.56	67.25	84.29	49.47	81.47	44.63	59.97	59.46	54.23	44.28	41.20
	CC3M	2	CLIP-Refine	61.10	62.71	88.46	66.71	83.96	47.80	80.76	44.36	59.21	<u>60.17</u>	54.60	43.67	40.75
	COCO Caption	1	ComCLIP	62.97	68.89	91.01	67.99	81.71	47.20	90.13	<u>46.81</u>	58.63	57.56	55.92	48.36	41.45
	CC3M	1	ComCLIP	63.15	69.31	91.50	68.51	82.70	49.55	89.67	46.81	59.02	55.76	53.24	48.95	<u>42.80</u>
	CC3M	2	ComCLIP	62.84	69.38	91.02	67.80	82.53	48.57	89.59	46.60	58.92	56.15	52.28	48.97	42.25
	CC12M	1	ComCLIP	<u>63.04</u>	<u>69.18</u>	<u>90.66</u>	<u>66.66</u>	81.94	47.67	89.37	46.12	59.11	58.74	54.49	<u>49.44</u>	43.10
ViT-L/14	—	-	OpenAI CLIP	66.12	75.54	95.19	71.05	83.27	49.96	93.24	55.27	63.33	62.65	52.01	59.59	32.35
	ImageNet-1k	2	KUEA	66.89	75.93	95.95	75.13	83.78	50.06	93.24	55.69	64.08	62.30	52.15	59.93	34.45
	CC3M	2	KUEA	66.36	75.53	95.53	72.44	83.50	50.03	93.35	55.37	63.57	62.67	52.02	59.70	32.65
	COCO Caption	1	CLIP-Refine	66.31	74.16	94.76	70.31	84.52	48.75	91.39	55.74	63.92	66.80	54.40	58.68	32.25
	COCO Caption	1	ComCLIP	67.09	75.82	95.18	74.07	82.96	52.72	93.35	55.90	63.56	66.74	52.79	59.95	32.05
	CC3M	1	ComCLIP	67.75	76.10	95.37	73.89	82.53	53.89	93.35	56.76	64.62	69.19	53.63	<u>60.25</u>	<u>33.45</u>
	CC3M	2	ComCLIP	67.78	76.11	95.25	73.99	82.73	54.07	<u>93.46</u>	56.91	64.46	68.80	53.98	60.20	33.35
	CC12M	1	ComCLIP	67.78	<u>75.81</u>	<u>95.64</u>	<u>74.67</u>	83.37	<u>53.78</u>	93.54	<u>55.69</u>	<u>64.54</u>	<u>67.57</u>	<u>54.84</u>	60.43	<u>33.45</u>
ViT-SO 400M/14	—	-	SigLIP	71.24	82.02	96.87	84.30	86.01	49.89	95.26	71.06	72.56	62.74	50.55	74.02	29.60
	ImageNet-1k	2	KUEA*	71.67	82.18	96.88	84.89	86.11	49.16	95.42	71.06	72.84	65.54	50.11	73.92	31.90
	CC3M	1	ComCLIP	72.15	82.50	97.31	85.14	<u>85.54</u>	<u>49.14</u>	95.26	71.91	<u>72.48</u>	<u>62.11</u>	60.33	<u>74.38</u>	<u>29.70</u>

(63.30 vs. 68.35 for the original CLIP), whereas ComCLIP varies by at most 0.2 across the three post-training datasets.

Table 15: Zero-shot image-text and text-to-image retrieval results on MSCOCO and Flickr30K benchmarks. The best and second-best results in each model block are highlighted in **bold** and underline, respectively.

Model	Dataset	epochs	Method	MSCOCO						Flickr30K					
				Image-to-Text			Text-to-Image			Image-to-Text			Text-to-Image		
				R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
ViT-B/16	—	-	OpenAI CLIP	48.16	72.58	81.88	31.47	55.62	66.73	74.70	92.90	96.40	57.10	81.36	88.20
	ImageNet-1k	2	KUEA	48.48	72.98	82.24	31.99	56.35	67.21	75.00	93.10	96.60	57.62	81.82	88.66
	CC3M	2	KUEA	48.00	72.64	81.76	31.47	55.79	66.71	75.00	92.80	96.60	57.00	81.46	88.38
	COCO Caption	1	CLIP-Refine	53.96	79.02	86.60	37.67	63.83	74.26	80.90	95.40	97.50	64.64	87.34	92.46
	CC3M	1	CLIP-Refine	41.12	67.54	77.88	33.29	58.50	69.12	66.20	91.30	95.40	60.18	84.64	90.64
	CC3M	2	CLIP-Refine	39.38	66.64	77.52	33.73	58.87	69.68	65.50	90.10	95.30	60.42	85.12	90.96
	COCO Caption	1	ComCLIP	<u>52.24</u>	<u>76.56</u>	<u>84.60</u>	35.36	61.09	71.78	<u>79.50</u>	94.70	96.70	61.36	84.62	90.80
	CC3M	1	ComCLIP	52.20	76.26	84.40	36.18	61.44	72.11	77.80	94.30	<u>97.40</u>	63.22	86.06	91.88
	CC3M	2	ComCLIP	52.02	75.86	84.06	<u>36.24</u>	<u>61.51</u>	<u>72.19</u>	77.20	94.20	96.90	<u>63.44</u>	<u>86.20</u>	91.84
	CC12M	1	ComCLIP	51.58	75.84	83.78	35.56	60.70	70.99	77.10	<u>94.80</u>	97.50	62.42	85.48	91.32
ViT-L/14	—	-	OpenAI CLIP	50.04	73.62	82.88	34.29	58.88	69.16	77.90	94.60	98.10	59.80	83.48	89.26
	ImageNet-1k	2	KUEA	50.88	75.38	83.80	35.66	60.47	70.64	79.50	95.30	98.30	61.18	84.70	90.24
	CC3M	2	KUEA	50.30	74.22	83.08	34.68	59.25	69.43	78.00	94.80	98.30	60.02	83.64	89.66
	COCO Caption	1	CLIP-Refine	53.28	77.42	85.30	38.21	63.05	73.24	80.10	96.20	98.30	64.56	87.86	<u>92.74</u>
	COCO Caption	1	ComCLIP	54.32	77.96	85.78	38.79	63.93	73.77	81.80	97.00	99.10	64.98	86.76	92.32
	CC3M	1	ComCLIP	<u>54.56</u>	77.96	85.98	39.40	64.45	74.12	82.60	96.30	98.80	65.82	88.22	92.62
	CC3M	2	ComCLIP	54.88	<u>77.46</u>	<u>85.82</u>	39.63	64.79	74.31	<u>82.10</u>	<u>96.30</u>	<u>98.90</u>	66.28	88.52	92.98
	CC12M	1	ComCLIP	53.74	77.30	85.38	38.63	63.73	73.35	80.10	95.30	98.40	65.00	87.74	92.14
ViT-SO 400M/14	—	-	SigLIP	68.16	87.46	92.86	52.20	75.71	83.37	88.90	98.50	99.60	74.80	92.24	95.22
	CC3M	1	ComCLIP	70.06	88.24	93.60	53.05	76.48	83.84	89.50	98.90	99.80	75.78	92.88	95.94

K Detailed Image-text Retrieval Experimental Results

Table 15 presents the detailed results (R@1, R@5, and R@10) of the baseline model and our proposed ComCLIP on two image-text retrieval datasets, covering various model architectures, training datasets, and training epochs.

On ViT-B/16, CLIP-Refine (COCO Caption) achieves the best retrieval performance on all metrics, and ComCLIP is second. On ViT-L/14, ComCLIP achieves higher retrieval than KUEA and CLIP-Refine (COCO Caption) in these single runs.

L Detailed Linear Probe Experimental Results

Table 16 presents detailed results for the baselines and our proposed ComCLIP across 5 linear probe tests, covering various model architectures, training datasets, and epochs.

On ViT-B/16, ComCLIP achieves higher average linear-probe accuracy than KUEA and CLIP-Refine on all three post-training datasets; the advantage over CLIP-Refine is significant over three seeds (Table 2) and holds under dataset-matched comparisons (Sec. 4.2). On ViT-L/14, KUEA (ImageNet-1K) achieves the best average, plausibly aided by the overlap between its post-training data and the ImageNet-1K probe; ComCLIP is second and varies by at most 1.3 across post-training datasets for one epoch.

M Detailed MMVP Experimental Results

Table 17 presents detailed results for the baselines and our proposed ComCLIP on MMVP benchmark, covering various model architectures, training datasets, and epochs.

MMVP contains 135 evaluated image pairs in total and only 15 pairs per category, so single-run per-category values move in steps of 6.67 points and should not be interpreted individually. The figures we report are the multi-seed means in Table 2: on ViT-B/16, ComCLIP (14.81 ± 2.97) and CLIP-Refine (16.05 ± 1.14) are indistinguishable; on ViT-L/14, ComCLIP (24.20 ± 0.86) is above CLIP-Refine trained on the same data (19.01 ± 1.86). Generative enhancement methods report higher MMVP on ViT-L/14 (Appendix N).

N Comparison with Generative-Model-Based Enhancement

DIVA (Wang et al., 2025), GenHancer (Ma et al., 2025) and un2CLIP (Li et al., 2025) improve CLIP’s fine-grained perception using feedback from a large generative model. A matched retraining

Table 16: Linear probe evaluation results on various datasets. The best and second-best results in each model block are highlighted in **bold** and underline, respectively.

Model	Dataset	epochs	Method	Avg	IN-1K	SVHN	GTSRB	CLEVR Dist.	CLEVR Counts
ViT-B/16	—	-	OpenAI CLIP	44.70	66.34	45.19	56.94	31.37	23.68
	IN-1k	2	KUEA	44.65	66.82	47.85	58.19	28.34	22.07
	CC3M	2	KUEA	44.72	66.46	46.22	57.48	30.83	22.62
	COCO Caption	1	CLIP-Refine	42.01	67.51	44.53	56.60	18.58	22.83
	CC3M	1	CLIP-Refine	44.89	66.22	45.61	56.94	31.38	24.28
	CC3M	2	CLIP-Refine	43.52	65.74	45.20	56.76	25.89	23.99
	COCO Caption	1	ComCLIP	47.23	67.00	46.27	57.13	29.49	36.25
	CC3M	1	ComCLIP	<u>48.74</u>	<u>67.15</u>	48.06	58.26	<u>32.69</u>	<u>37.53</u>
	CC3M	2	ComCLIP	49.98	66.62	46.96	57.66	38.11	40.55
	CC12M	1	ComCLIP	46.56	66.69	45.86	57.09	26.38	36.79
ViT-L/14	—	-	OpenAI CLIP	54.90	72.07	65.21	72.96	22.97	41.29
	IN-1k	2	KUEA	59.92	73.37	71.04	<u>73.77</u>	28.99	52.42
	CC3M	2	KUEA	55.82	72.45	66.79	73.59	23.89	42.37
	COCO Caption	1	CLIP-Refine	50.83	72.84	64.11	72.84	21.10	23.28
	COCO Caption	1	ComCLIP	<u>58.28</u>	72.11	66.71	74.46	<u>26.43</u>	51.69
	CC3M	1	ComCLIP	56.99	<u>72.85</u>	64.78	73.53	21.49	<u>52.29</u>
	CC3M	2	ComCLIP	56.60	72.63	64.65	73.20	21.03	51.47
	CC12M	1	ComCLIP	57.42	72.71	65.50	73.65	24.25	50.99
ViT-SO	—	-	SigLIP	70.87	80.78	78.02	89.63	37.94	67.98
400M/14	CC3M	1	ComCLIP	70.88	80.84	78.02	89.63	37.93	67.99

Table 17: Detailed MMVP benchmark results (single runs; each category has only 15 pairs, so one pair corresponds to 6.67 points). [‡]: per-category breakdown of one seed; on ViT-B/16 this seed is the maximum of three (3-seed mean 14.81 ± 2.97), and on ViT-L/14 it is one of three seeds (3-seed mean 24.20 ± 0.86). The multi-seed means in Table 2 are the figures we report; per-category and single-run differences are not used for any conclusion. The best results in each model block are highlighted in **bold**.

Model	Dataset	epochs	Method	Avg	Orientation and	Presence of Specific	State and Condition	Quantity and	Positional and	Color and Appearance	Structural and	Texts	Viewpoint and Perspective
ViT-B/16	—	-	OpenAI CLIP	12.59	6.67	0.00	26.67	13.33	13.33	20.00	13.33	0.00	20.00
	ImageNet-1k	2	KUEA	11.85	0.00	0.00	20.00	13.33	13.33	26.67	13.33	0.00	20.00
	CC3M	2	KUEA	14.07	6.67	0.00	26.67	20.00	13.33	20.00	20.00	0.00	20.00
	COCO Caption	1	CLIP-Refine	17.04	13.33	6.67	20.00	0.00	20.00	46.67	33.33	0.00	13.33
	CC3M	1	CLIP-Refine	19.26	13.33	0.00	26.67	6.67	26.67	53.33	33.33	0.00	13.33
	CC3M	2	CLIP-Refine	20.74	13.33	0.00	33.33	20.00	20.00	40.00	33.33	6.67	20.00
	COCO Caption	1	ComCLIP	15.56	13.33	6.67	33.33	6.67	13.33	20.00	33.33	0.00	13.33
	CC3M	1	ComCLIP [‡]	17.78	6.67	0.00	26.67	0.00	13.33	46.67	33.33	6.67	26.67
	CC3M	2	ComCLIP	16.30	6.67	0.00	33.33	0.00	13.33	46.67	33.33	6.67	6.67
	CC12M	1	ComCLIP	15.56	6.67	0.00	33.33	0.00	6.67	26.67	40.00	6.67	20.00
ViT-L/14	—	-	OpenAI CLIP	17.78	6.67	13.33	20.00	13.33	6.67	53.33	26.67	6.67	13.33
	ImageNet-1k	2	KUEA	17.78	6.67	20.00	20.00	6.67	6.67	40.00	33.33	6.67	20.00
	CC3M	2	KUEA	18.52	13.33	13.33	20.00	13.33	6.67	46.67	20.00	13.33	20.00
	COCO Caption	1	CLIP-Refine	17.04	6.67	13.33	26.67	13.33	6.67	33.33	13.33	13.33	26.67
	COCO Caption	1	ComCLIP	20.74	6.67	13.33	33.33	13.33	6.67	53.33	13.33	13.33	33.33
	CC3M	1	ComCLIP [‡]	25.19	20.00	20.00	33.33	13.33	6.67	60.00	20.00	13.33	40.00
	CC3M	2	ComCLIP	25.93	20.00	26.67	40.00	13.33	6.67	40.00	26.67	13.33	46.67
	CC12M	1	ComCLIP	22.22	20.00	13.33	26.67	13.33	6.67	46.67	26.67	20.00	26.67
ViT-SO	—	-	SigLIP	39.26	26.67	20.00	53.33	40.00	20.00	66.67	40.00	26.67	60.00
400M/14	CC3M	1	ComCLIP	41.48	26.67	33.33	60.00	40.00	13.33	66.67	46.67	26.67	60.00

Table 18: Comparison with generative-model-based enhancement methods on OpenAI CLIP ViT-L/14 (224px), using the numbers *published* by each paper (base $\rightarrow$ post-trained, as reported in that paper). Evaluation protocols and base values differ across papers (e.g., retrieval R@1 base values), so the rows are not strictly comparable; each row should be read as the change relative to its own base. “—”: not reported in the form used here. [†] A third-party reproduction from GenHancer’s official checkpoints reports ImageNet-1K 40.2 (Li et al., 2025); the two sources disagree. ComCLIP MMVP is the 3-seed mean (Table 2).

Method	Enhancement signal	IN-1K zero-shot	Retrieval (as reported)	MMVP-VLM
DIVA (Wang et al., 2025)	diffusion feedback	75.5 $\rightarrow$ 75.5	COCO R@1 I $\rightarrow$ T 56.4 $\rightarrow$ 56.7, T $\rightarrow$ I 36.5 $\rightarrow$ 36.6	19.3 $\rightarrow$ 25.9
GenHancer (Ma et al., 2025)	generative reconstruction	73.3 $\rightarrow$ 75.6 [†]	COCO text retrieval R@5 79.2 $\rightarrow$ 79.4	31.9
un2CLIP (Li et al., 2025)	inverted unCLIP	75.5 $\rightarrow$ 62.4	COCO T $\rightarrow$ I R@5 61.0 $\rightarrow$ 65.5; Flickr T $\rightarrow$ I R@5 87.3 $\rightarrow$ 90.1	32.6
ComCLIP (ours, CC3M)	contrastive + anchor + RKD, 1 epoch	75.54 $\rightarrow$ 76.10	COCO R@1 I $\rightarrow$ T 50.04 $\rightarrow$ 54.56, T $\rightarrow$ I 34.29 $\rightarrow$ 39.40; Flickr T $\rightarrow$ I R@5 83.48 $\rightarrow$ 88.22	17.78 $\rightarrow$ 24.20 $_{\pm 0.86}$

is outside our compute class, so Table 18 reports their published numbers on OpenAI CLIP ViT-L/14. Contrary to a common characterization (including our own submitted version), these methods do not in general sacrifice CLIP’s other capabilities: DIVA reports unchanged zero-shot classification (27-dataset average $69.3 \rightarrow 69.3$; ImageNet-1K $75.5 \rightarrow 75.5$) with slightly improved COCO retrieval, and GenHancer reports changes under 0.3 points on ImageNet-1K, SUN397 and COCO retrieval (a third-party reproduction from its official checkpoints reports a large ImageNet drop, and we

cannot resolve the disagreement). un2CLIP reduces ImageNet-1K accuracy ($75.5 \rightarrow 62.4$) while improving text-to-image retrieval. On MMVP, GenHancer (31.9) and un2CLIP (32.6) are well above ComCLIP (24.20 ± 0.86), and DIVA (25.9) is comparable, although base values differ across papers. ComCLIP is therefore not a method for maximizing fine-grained perception. It occupies a different point: a single-epoch contrastive refinement at a fixed text encoder, without a generative model in the loop, whose supported contributions are the temperature diagnosis and improved linear-probe transferability.

O Broader Impacts

ComCLIP is a lightweight post-training method that improves an existing vision-language encoder with one epoch on public data, which lowers the compute and energy cost of refining CLIP compared with re-training from scratch. Because the text encoder is frozen, ComCLIP can be dropped into existing CLIP-based systems such as LLaVA without re-training them; more transferable visual features may benefit downstream applications such as accessibility tools and visual question answering. On the other hand, ComCLIP inherits the biases of the pretrained CLIP and DINOv2 models and of the web-scraped post-training data (CC3M, CC12M), and our post-training does not explicitly mitigate them. Improved visual encoders could also be misused, for example in surveillance applications. We encourage practitioners to evaluate downstream systems built on ComCLIP for fairness and safety before deployment.

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